MODULE 5 NERVE PLEXUSES

The Nerve Plexuses.

THE PERIPHERAL NERVOUS SYSTEM, THE NERVE PLEXUSES

A brief reference for the nerve plexuses. A plexus is a network where the anterior rami of spinal nerves regroup before traveling to the body. Covers the principal plexuses at a general level: where each one forms and what region it serves, so you can predict the area of the body affected if a plexus is damaged.

BY THE END

1. Define a nerve plexus and explain why the anterior rami regroup.
2. Name the principal nerve plexuses and the spinal nerves that form each.
3. Predict the region of the body affected if a given plexus is damaged.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The nerve plexuses

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

What a Nerve Plexus Is

Most spinal nerves do not run straight to their targets. The anterior rami of neighboring spinal nerves first weave together into a network called a plexus, then sort their axons into the nerves that leave it. The thoracic region is the exception.

THE PLEXUS AND WHY IT EXISTS

Structure	What it is
Nerve plexus	A network where the anterior rami of several spinal nerves join and regroup before their axons travel on to the body
Anterior rami	The branches of the spinal nerves that form the plexuses and supply the limbs and the front of the body
Why a plexus forms	Regrouping lets each body region receive axons from several spinal nerves, so damage to one spinal nerve does not erase all function in that area
Intercostal nerves	The exception; the anterior rami of T2 to T12 do not form a plexus but run directly to the muscles and skin of the thoracic wall
Principal plexuses	The cervical, brachial, lumbar, sacral, and coccygeal plexuses, each named for the region it serves

The Principal Nerve Plexuses

Each plexus is formed by a known set of spinal nerves and serves a known region of the body. Compare them.

THE FIVE PLEXUSES, WHAT FORMS EACH, AND THE REGION IT SERVES

Plexus	Formed by	Region served
Cervical plexus	The anterior rami of C1 to C4, with a contribution from C5	The skin and muscles of the neck and the upper shoulders; it also gives rise to the phrenic nerve, the motor supply to the diaphragm
Brachial plexus	The anterior rami of C5 to C8 and T1	The shoulder and the entire upper limb
Lumbar plexus	The anterior rami of L1 to L4	The anterolateral abdominal wall, the external genitals, and part of the lower limb
Sacral plexus	The anterior rami of L4 to L5 and S1 to S4	The buttocks, the perineum, and most of the lower limb
Coccygeal plexus	The anterior rami of S4 to S5 and the coccygeal nerves	A small area of skin over the coccyx

KEY POINT

The plexuses run head to tail in the same order as the cord regions that build them: cervical for the neck, brachial for the upper limb, lumbar for the abdominal wall and thigh, sacral for the buttock and lower limb, coccygeal for the skin over the coccyx.

Predicting the Effect of Plexus Damage

Because each plexus serves one region, the location of a plexus injury predicts the region of the body that loses movement and sensation. This is the level of detail to know for this course.

INJURY SITE AND THE REGION AFFECTED

Plexus damaged	Region affected
Cervical plexus	Weakness or numbness of the neck and shoulder; if the phrenic nerve is involved, the diaphragm on that side is weakened and breathing is affected
Brachial plexus	Weakness and sensory loss across the shoulder and the whole upper limb on that side
Lumbar plexus	Weakness and sensory loss of the lower abdominal wall and the front and inner surfaces of the thigh
Sacral plexus	Weakness and sensory loss of the buttock, the back of the thigh, and the leg and foot
Coccygeal plexus	Numbness of the small patch of skin over the coccyx

WHY THE NETWORK MATTERS

The plexus design is a safety feature. Because each limb region draws axons from several spinal nerves, a single damaged spinal nerve root weakens an area but rarely paralyzes it completely. The overlap covers the gap.

Damage to the **plexus** itself is different. A plexus injury sits downstream of that regrouping, so it can take out a whole region at once. The level of the injury predicts the deficit: a brachial plexus injury affects the upper limb, a lumbar or sacral injury the lower limb.

The **phrenic nerve** is the one to remember. It leaves the cervical plexus and is the sole motor supply to the diaphragm. The teaching phrase "C3, 4, 5 keep the diaphragm alive" is a reminder that a high cervical injury can stop breathing.